B.Tech (ME)– IV-I Sem

23A03702d	MECHATRONICS AND MEMS (Professional Elective-IV)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	**To introduce** the concept of mechatronics and its significance in modern industrial applications.
2	**To explain** the working principles, characteristics, and selection of various types of sensors.
3	**To describe** different actuator systems and their integration in automation circuits.
4	**To develop** understanding of microprocessors, microcontrollers, and PLCs including programming and control logic.
5	**To explore** MEMS technologies, fabrication techniques, and their applications in miniaturized systems.

Course Outcomes: On successful completion of the course, the student will be able to,		
1	Define and explain the fundamental concepts and applications of mechatronics in industrial systems.	L1,L2, L3
2	Identify and analyze various types of sensors and their performance characteristics for specific applications.	L2,L4,L5
3	Design and evaluate actuator systems including hydraulic and pneumatic circuits for automation tasks.	L3,L5,L6
4	Understand and develop PLC ladder logic programs for industrial automation using sequencing, timers, counters, and analog I/O.	L2,L3,L6
5	Describe MEMS fabrication methods, apply and analyze the effect of scaling in micro electromechanical systems.	L2,L3,L4

UNIT I

Introduction

Definition of Mechatronics, Need for Mechatronics in Industry, Objectives of mechatronics, mechatronics design process, Mechatronics key elements, mechatronics applications – Computer numerical control (CNC) machines, Tool monitoring systems, Flexible manufacturing system (FMS), Industrial Robots, Automatic packaging systems, Automatic inspection systems.

UNIT II

Sensors

Static and dynamic characteristics of sensors, Displacement, Position and Proximity sensors, Force and torque sensors, Pressure sensors, Flow sensors, Temperature sensors, Acceleration sensors, Level sensors, Light sensors, Smart material sensors, Micro and Nano sensors, Selection criteria for sensors.

UNIT III

Actuators

Mechanical, Electrical, Hydraulic and Pneumatic Actuation systems, Characteristics and their limitations, Design of Hydraulic and Pneumatic circuits, Piezoelectric actuators, Shape memory alloys, Selection criteria for actuators.

UNIT IV

Microprocessors, Microcontrollers and Programmable Logic Controllers

Architecture of Microprocessor, Microcontroller and Programmable Logic Controller, PLC Programming using ladder diagrams, logics, latching, sequencing, timers relays and counters, data handling, Analog input/output, selection of controllers.

UNIT V

Micro Electro Mechanical Systems (MEMS)

History, Effect of scaling, Fabrication Techniques: Oxidation, Physical Vapor disposition, Chemical Vapor Deposition, Lithography, Etching, Wafer bonding, LIGA, DRIE, Applications: Lab on chip.

Textbooks:

1. Bolton, Mechatronics Electronics Control Systems in Mechanical and Electrical Engineering, 3/e, Pearson Education Press, 2005.
2. Devadas Shetty and Richard A Kolk, Mechatronic System Design, 2/e, Cengage learning, 2010.
3. N. Mahalik, MEMS, McGraw Hill Educations, 2017.

Reference Books:

1. Clarence W. de Silva, Mechatronics an Integrated Approach, CRC Press, 2004.
2. James J Allen, Micro Electro Mechanical Systems Design, CRC Press Taylor & Francis group, 2005.
3. Ganesh S Hedge, Mechatronics, Jones & Bartlett Learning, 2010.
4. Mohammed Gad, MEMS; Design and Fabrication, CRC Press, 2010.

Online Learning Resources:

- https://onlinecourses.nptel.ac.in/noc22_me54/preview
- https://nptel.ac.in/courses/112108092
- https://nptel.ac.in/courses/112101304
- https://onlinecourses.nptel.ac.in/noc20_ee56/preview
- https://www.cet.edu.in/noticefiles/259_Lecturer%20Note%20on%20Mechatronics-ilovepdf-compressed.pdf, https://lecturenotes.in/subject/1176/mechatronics-and-mems